

Q.4 Root locus technique is used to check the stability of a control system (True/False)

Q.5 Laplace transform of Step Signal is.

- a) S b) $1/S$
c) S^2 d) $1/S^2$

Q.6 The type-2 system has numbers of pole at origin.

- a) 1 b) 2
c) 3 d) 4

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define the Transfer function.

Q.8 Explain / Write down the Mason's gain formula.

Q.9 Write any three standard test signals.

Q.10 Define Steady state error.

Q.11 Define Backlash.

Q.12 Define Dead zone.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

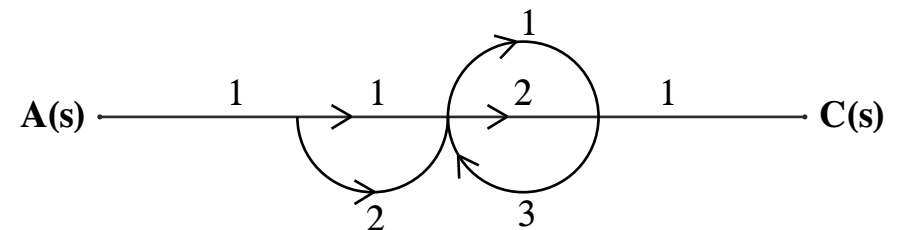
Q.13 Write down the five differences between OLCS and CLCS.

Q.14 Explain the all basic elements of control system.

Q.15 Draw the Mathematical modeling of mechanical translational systems.

Q.16 Define Laplace transform I find out the Laplace transforms of unit ramp signal.

Q.17 Find out the transfer function of given signal flow graph.



Q.18 Describe all standard test signal. Used in time response analysis.

Q.19 Find the stability for given characteristics eq"

$$S^4 + 3S^3 + 25S^2 + 5S + 2 = 0$$